

# **NORTHWIND SALES PERFORMANCE DASHBOARD**

## **Using Microsoft Power BI**

### **Project Report**

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#### **1. Introduction**

Power BI is a business intelligence and data visualization tool developed by Microsoft. It helps users convert raw data into meaningful reports, interactive dashboards and visualizations.

For this project, a **Northwind sales dataset** was used to create an interactive Sales Dashboard using Microsoft Power BI.

The dashboard presents information about orders, customers, products, categories, employees, sales, quantity sold and freight.

The main purpose of the dashboard is to analyse sales information and present it through KPI cards, charts, tables and interactive filters.

#### **2. Objectives of the Project**

The major objectives of this project are:

- To analyse the sales data using Microsoft Power BI.
- To create an interactive sales dashboard.
- To analyse the total number of orders.
- To analyse the total quantity sold.
- To analyse total freight and average freight.
- To analyse sales across different product categories.
- To analyse order trends by month.
- To compare orders handled by employees.
- To analyse customers by country.
- To analyse individual product sales and quantity sold.
- To provide interactive filtering using Year, Country and Category.

### 3. Dataset Description

The dataset used for this project contains Northwind sales information.

The dashboard displays information related to:

| Field               | Description                                       |
|---------------------|---------------------------------------------------|
| Product Name        | Name of the product                               |
| Category Name       | Category of the product                           |
| Total Sales         | Sales amount associated with the product/category |
| Total Quantity Sold | Quantity of products sold                         |
| Employee Name       | Employee associated with orders                   |
| Country             | Customer country                                  |
| Year                | Year used for filtering                           |
| Freight             | Freight associated with orders                    |
| Orders              | Number of orders                                  |

The dashboard contains product categories such as **Beverages, Dairy Products, Confections, Meat & Poultry, Seafood, Condiments, Produce, and Grains & Cereals.**

### 4. Data Preparation

Before creating the dashboard, the data was prepared in Microsoft Power BI.

The main preparation activities included:

- Importing the Northwind data into Power BI.
- Preparing the product and category information.
- Preparing customer country information.
- Preparing employee information.
- Preparing order information.

- Preparing sales and quantity fields for analysis.
- Preparing date information for monthly analysis.
- Creating measures required for the KPI cards and visualizations.

The prepared data was then used to create different dashboard pages and visualizations.

## **5. DAX Measures Used**

The dashboard contains measures for calculating the main business metrics.

### **→ Total Orders**

This measure represents the total number of orders displayed in the dashboard.

**Displayed value: 809**

#### **Purpose:**

- To understand the overall order volume.
- To provide a quick view of the number of orders.

### **→ Number of Customers**

This measure represents the number of customers displayed in the dashboard.

**Displayed value: 91**

#### **Purpose:**

- To understand the customer count.
- To support customer analysis by country.

### **→ Total Quantity Sold**

This measure represents the total quantity of products sold.

**Displayed value: 51K**

#### **Purpose:**

- To understand the quantity of products sold.
- To provide an overall sales-volume indicator.

### → Total Freight

This measure represents the total freight amount.

**Displayed value: 63.96K**

#### **Purpose:**

- To understand the total freight associated with the orders.
- To provide information about freight costs.

### → Average Freight

This measure represents the average freight value.

**Displayed value: 79.05**

#### **Purpose:**

- To understand the average freight associated with orders.

## **6. Dashboard Components**

The Week 3 Power BI project contains multiple dashboard pages.

The main pages shown in the project are:

1. **Sales Over View**
2. **Product and Customer Analysis**
3. **Interactive Sales Board**
4. **Category Tooltip**
5. **Category Details**

Each page provides a different view of the Northwind sales information.

## **7. KPI Cards**

The Sales Overview dashboard contains KPI cards displaying the main metrics.

The cards show:

- **Total Orders – 809**

- **Number of Customers – 91**
- **Total Quantity Sold – 51K**
- **Total Freight – 63.96K**
- **Average Freight – 79.05**

These cards provide a quick summary of the information available in the dashboard.

## 8. Total Orders by Month

A **Line Chart** is used to display Total Orders by Month.

The chart contains:

- January
- February
- March
- April
- May
- June
- July
- August
- September
- October
- November
- December

The line chart shows that the number of orders changes across the months.

From the displayed chart, the order count rises around **March**, decreases considerably during **May and June**, and increases again toward **December**.

### Purpose:

- To analyse monthly order trends.
- To compare order activity between months.
- To identify months with relatively higher or lower displayed order activity.

## 9. Total Sales by Category

A **Column Chart** is used to display Total Sales by Category.

The categories displayed are:

- Beverages
- Dairy Products
- Confections
- Meat & Poultry
- Seafood
- Condiments
- Produce
- Grains & Cereals

The chart shows differences in sales contribution between the product categories.

The displayed chart shows **Beverages** with the tallest sales column, followed by **Dairy Products**.

### Purpose:

- To compare sales between categories.
- To understand category-level sales contribution.
- To visually analyse differences between product categories.

## 10. Total Orders by Employee

A **Horizontal Bar Chart** is used to display Total Orders by Employee.

The employees displayed include:

- Margaret Peacock
- Janet Leverling
- Nancy Davolio
- Laura Callahan
- Robert King

- Michael Suyama
- Anne Dodsworth
- Steven Buchanan

The chart also contains a **(Blank)** category.

The visualization allows the order counts associated with the displayed employees to be compared.

**Purpose:**

- To analyse orders by employee.
- To compare employee-related order activity.
- To understand the distribution of orders across employees.

## 11. Product and Customer Analysis

The **Product and Customer Analysis** page contains a detailed product table.

The table displays:

- Product Name
- Category Name
- Total Sales
- Total Quantity Sold

Examples of products visible in the table include:

- Chai
- Chang
- Chartreuse verte
- Côte de Blaye
- Guarana Fantastica
- Ipoh Coffee
- Lakkalikööri
- Laughing Lumberjack Lager
- Outback Lager

- Rhönbräu Klosterbier
- Sasquatch Ale

The table provides detailed information about individual products.

**Purpose:**

- To analyse individual products.
- To compare product sales.
- To compare quantities sold.
- To understand product-level performance.

## **12. Number of Customers by Country**

A **Horizontal Bar Chart** is used to display Number of Customers by Country.

Countries visible in the chart include:

- USA
- France
- Germany
- Brazil
- UK
- Mexico
- Spain
- Venezuela
- Argentina
- Canada
- Italy
- Austria
- Belgium
- Denmark
- Finland

- Portugal
- Sweden
- Switzerland
- Ireland
- Norway
- Poland

The chart visually compares the number of customers across these countries.

The displayed chart shows the **USA** with the longest bar, followed by **France and Germany**.

**Purpose:**

- To analyse customer distribution by country.
- To compare customer counts between countries.
- To understand the geographical distribution of customers.

### **13. Interactive Sales Board**

The **Interactive Sales Board** combines multiple visualizations and filters into one dashboard.

The dashboard contains:

- Total Orders by Month
- Total Orders by Employee
- Total Sales by Category
- Year slicer
- Country slicer
- Category Name slicer
- KPI cards

The Year filter visibly contains:

- 2013
- 2014

The Country and Category Name filters can also be used to filter the dashboard.

**Purpose:**

- To provide interactive analysis.
- To allow users to filter the dashboard.
- To compare information across years.
- To analyse selected countries.
- To analyse selected product categories.

#### **14. Category Tooltip**

The **Category Tooltip** page provides a compact view of selected category information.

The screenshot displays three KPI cards:

**Total Sales**

**267.87K**

**Total Quantity Sold**

**10K**

**Total Orders**

**809**

These values are presented in the tooltip view shown in the dashboard.

**Purpose:**

- To display category information in a compact format.
- To provide additional information when analysing categories.
- To make the dashboard more interactive.

#### **15. Category Details**

The **Category Details** page provides detailed product information.

The table contains:

- Product Name
- Quantity Per Unit

- Category Name
- Total Sales
- Total Quantity Sold

The displayed products include products from the **Beverages** category.

Examples visible in the table include:

- Chai
- Guarana Fantastica
- Côte de Blaye
- Ipoh Coffee
- Rhönbräu Klosterbier
- Chang
- Laughing Lumberjack Lager
- Sasquatch Ale

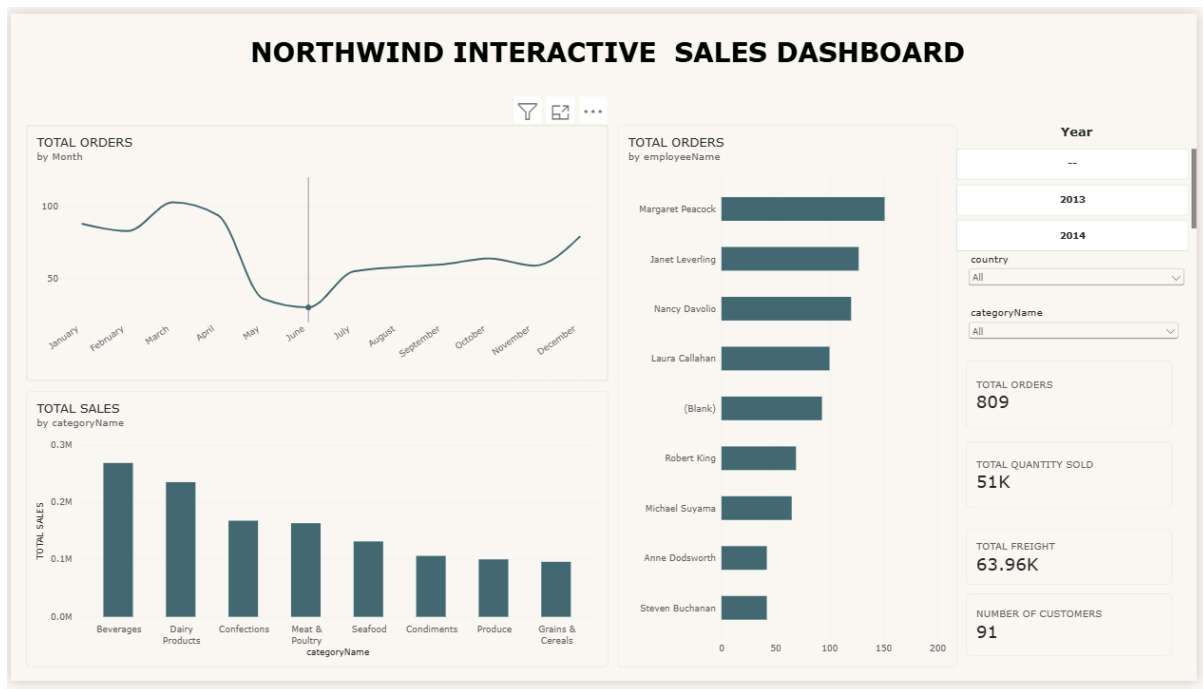
The table also contains a **Total** row for the displayed data.

**Purpose:**

- To analyse individual products within a category.
- To compare product sales.
- To compare quantity sold.
- To provide detailed category-level information.

## 16. Slicers Used

The Interactive Sales Board contains the following slicers:



### 1. Year

The Year slicer allows the user to select a year.

The displayed options include **2013** and **2014**.

### 2. Country

The Country slicer allows the dashboard to be filtered according to country.

### 3. Category Name

The Category Name slicer allows the user to select a product category.

These slicers make the dashboard interactive and allow the displayed information to change according to the selected filters.

## 17. Key Findings

Based strictly on the information visible in the Week 3 dashboard:

- The dashboard displays **809 total orders**.
- The dashboard displays **91 customers**.
- The dashboard displays approximately **51K total quantity sold**.
- The dashboard displays **63.96K total freight**.
- The dashboard displays **79.05 average freight**.

- The overall dashboard displays approximately **1.27M total sales**.
- The monthly order chart shows variation in order volume throughout the year.
- The category sales chart shows different sales contributions across the eight displayed categories.
- The employee chart shows differences in order counts between employees.
- The customer chart shows customers distributed across multiple countries.
- The product analysis page provides product-level sales and quantity information.
- The dashboard provides Year, Country and Category Name filters for interactive analysis.

**No additional numerical findings are being added where the screenshots do not provide an exact value.**

## **18. Importance of the Dashboard**

The Northwind Sales Dashboard provides several useful functions for analysing sales information.

### **Easy Data Understanding**

The dashboard converts sales information into charts, KPI cards and tables.

### **Quick Overview**

The KPI cards provide important information such as total orders, customers, quantity sold and freight.

### **Category Analysis**

The category chart allows sales across different categories to be compared.

### **Employee Analysis**

The employee chart allows order activity to be compared between employees.

### **Customer Analysis**

The country chart provides a visual representation of customer distribution.

### **Product Analysis**

The product tables provide detailed information about individual products.

### **Interactive Analysis**

The Year, Country and Category Name slicers allow users to filter the dashboard.

## 19. Conclusion

The **Northwind Sales Performance Dashboard** demonstrates how Microsoft Power BI can be used to transform sales data into an interactive business dashboard.

The project uses KPI cards, line charts, bar charts, column charts, tables, slicers and tooltip pages to present different aspects of the Northwind sales information.

The dashboard provides information about **orders, customers, quantity sold, freight, sales categories, employees, countries and individual products**.

The project also demonstrates the use of interactive filtering through **Year, Country and Category Name** slicers.

Overall, the dashboard provides a structured visual representation of the available Northwind sales data and allows users to explore the information at both summary and detailed product levels.